RELATIVE PERFECT COMPLEXES ON A DIRECTED LIMIT OF SPACES DESCEND TO A FINITE STAGE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 12498-12503 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 12505-12575, the complete author proof (69 body lines).

Editorial note. Navigation stubs: 6 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The card's statement is relative to the chapter's Situation for relative descent of perfect complexes.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-descend-relatively-perfect' also exists in stacks-perfect — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Full faithfulness of the colimit comparison (same chapter, lemma-relative-descend-homomorphisms), the induction principle for bootstrap on derived categories of spaces (Derived Categories of Spaces, lemma-induction-principle), the affine-local characterization of relative perfectness (lemma-affine-locally-rel-perfect), and the schemes case (Derived Categories of Schemes, lemma-descend-relatively-perfect).

Source reference. Original lemma 0DKQ: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12313–12335. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DKV: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12435–12449. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DI8: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12575–12575. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DKU: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12421–12432. Cited by the retained statement or

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Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08HB: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12575–12575. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 08GP: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 12575–12575. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Situation 0DKU the category of Y-perfect objects of $D(\mathcal{O}_X)$ is the colimit of the categories of Y_i-perfect objects of $D(\mathcal{O}_{X_i})$.*

Proof. For every quasi-compact and quasi-separated object U_0 of $(X_0)_{spaces,étale}$ consider the condition P that the functor

$$\mathrm{colim}_{i\geq 0} D_{Y_i\text{-perfect}}(\mathcal{O}_{U_i}) \longrightarrow D_{Y\text{-perfect}}(\mathcal{O}_U)$$

is an equivalence where $U = X \times_{X_0} U_0$ and $U_i = X_i \times_{X_0} U_0$. We observe that we already know this functor is fully faithful by Lemma 0DKV. Thus it suffices to prove essential surjectivity.

Suppose that $(U_0 \subset W_0, V_0 \to W_0)$ is an elementary distinguished square in $(X_0)_{spaces,étale}$ and P holds for $U_0, V_0, U_0 \times_{W_0} V_0$. We claim that P holds for W_0. We will use the notation $U_i = X_i \times_{X_0} U_0$, $U = X \times_{X_0} U_0$, and similarly for V_0 and W_0. We will abusively use the symbol f_i for all the morphisms $U \to U_i$, $V \to V_i$, $U \times_W V \to U_i \times_{W_i} V_i$, and $W \to W_i$. Suppose E is an Y-perfect object of $D(\mathcal{O}_W)$. Goal: show E is in the essential image of the functor. By assumption, we can find $i \geq 0$, an Y_i-perfect object $E_{U,i}$ on U_i, an Y_i-perfect object $E_{V,i}$ on V_i, and isomorphisms $Lf_i^* E_{U,i} \to E|_U$ and $Lf_i^* E_{V,i} \to E|_V$. Let

$$a : E_{U,i} \to (Rf_{i,*}E)|_{U_i} \quad \text{and} \quad b : E_{V,i} \to (Rf_{i,*}E)|_{V_i}$$

the maps adjoint to the isomorphisms $Lf_i^* E_{U,i} \to E|_U$ and $Lf_i^* E_{V,i} \to E|_V$. By fully faithfulness, after increasing i, we can find an isomorphism $c : E_{U,i}|_{U_i \times_{W_i} V_i} \to E_{V,i}|_{U_i \times_{W_i} V_i}$ which pulls back to the identifications

$$Lf_i^* E_{U,i}|_{U\times_W V} \to E|_{U\times_W V} \to Lf_i^* E_{V,i}|_{U\times_W V}.$$

Apply Derived Categories of Spaces, Lemma 08HB to get an object E_i on W_i and a map $d : E_i \to Rf_{i,*}E$ which restricts to the maps a and b over U_i and V_i. Then it is clear that E_i is Y_i-perfect (because being relatively perfect is an étale local property) and that d is adjoint to an isomorphism $Lf_i^* E_i \to E$.

By exactly the same argument as used in the proof of Lemma 0DKV using the induction principle (Derived Categories of Spaces, Lemma 08GP) we reduce to the case where both X_0 and Y_0 are affine: first work with quasi-compact and quasi-separated objects in $(Y_0)_{spaces,étale}$ to reduce to Y_0 affine, then work with quasi-compact and quasi-separated object in $(X_0)_{spaces,étale}$ to reduce to X_0 affine. In the affine case the result follows from the case of schemes which is Derived Categories of Schemes, Lemma 0DI8. The translation into the case for schemes is done by Lemma 0DKQ. □